

Supplementary material:

Blood–brain barrier disruption in patients with persistent
post-concussion symptoms

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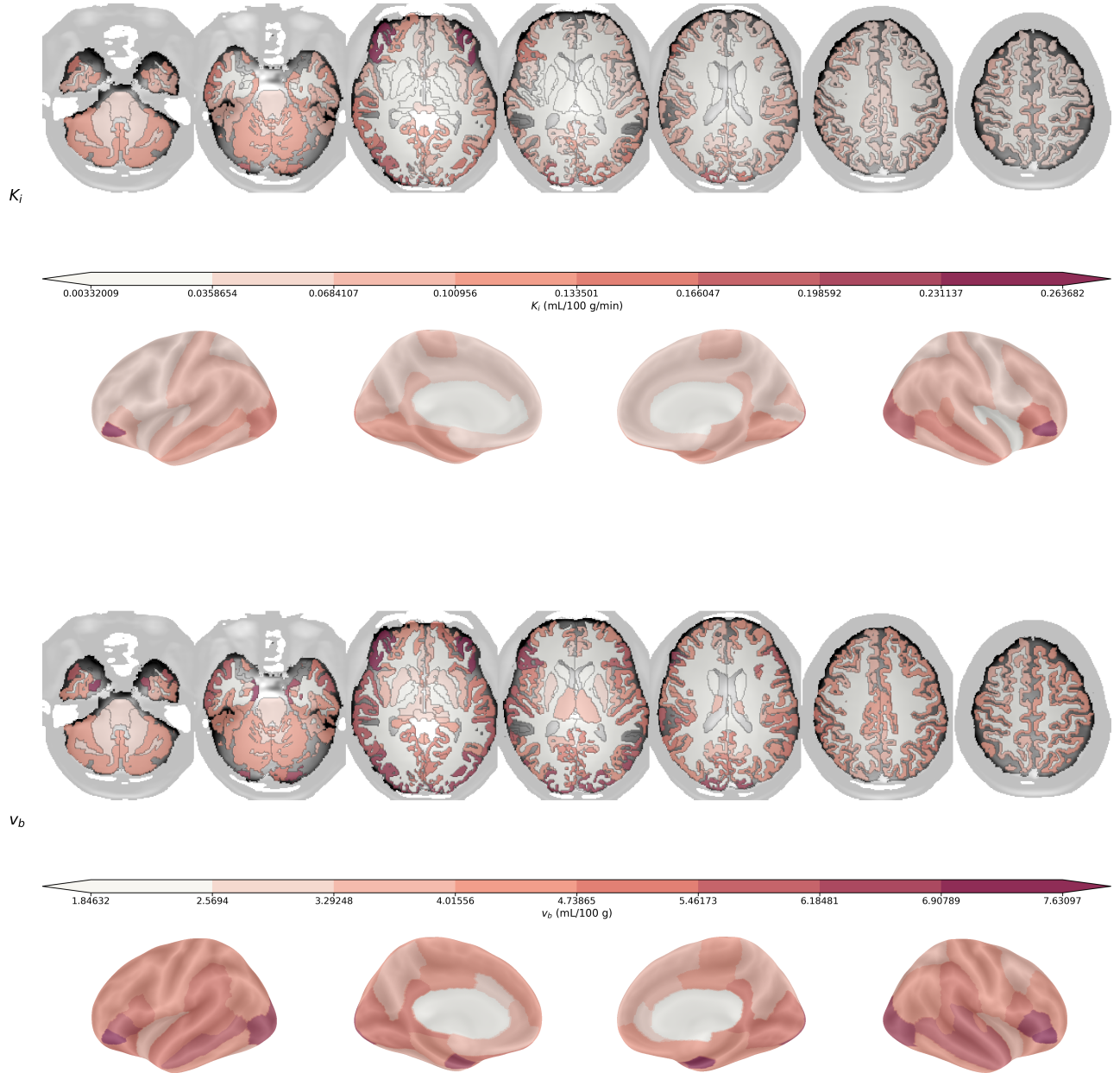
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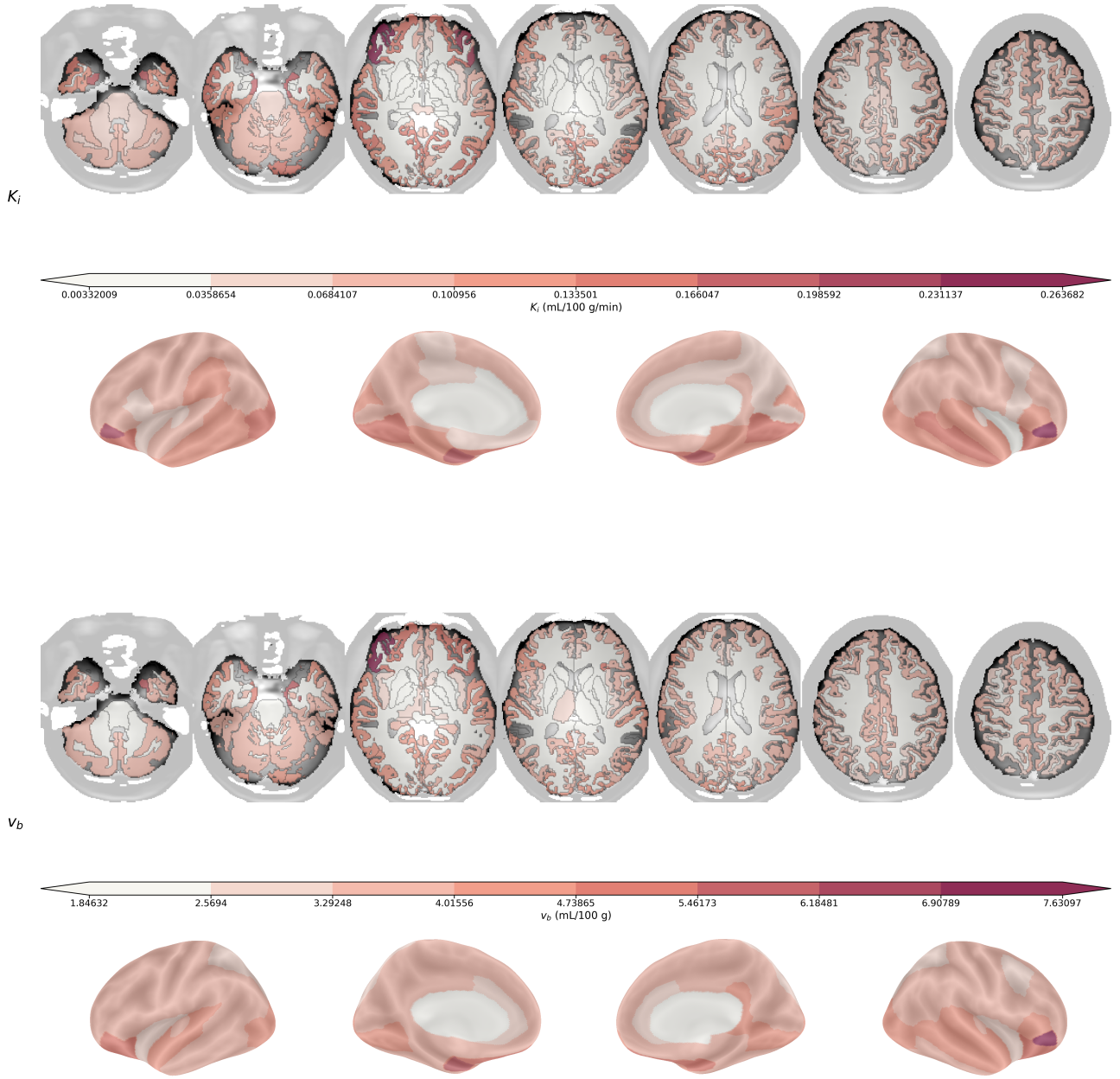
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Supplementary Figure S1: Group-mean parcel maps

Group-mean parcel-level values - Healthy controls ($n = 15$)

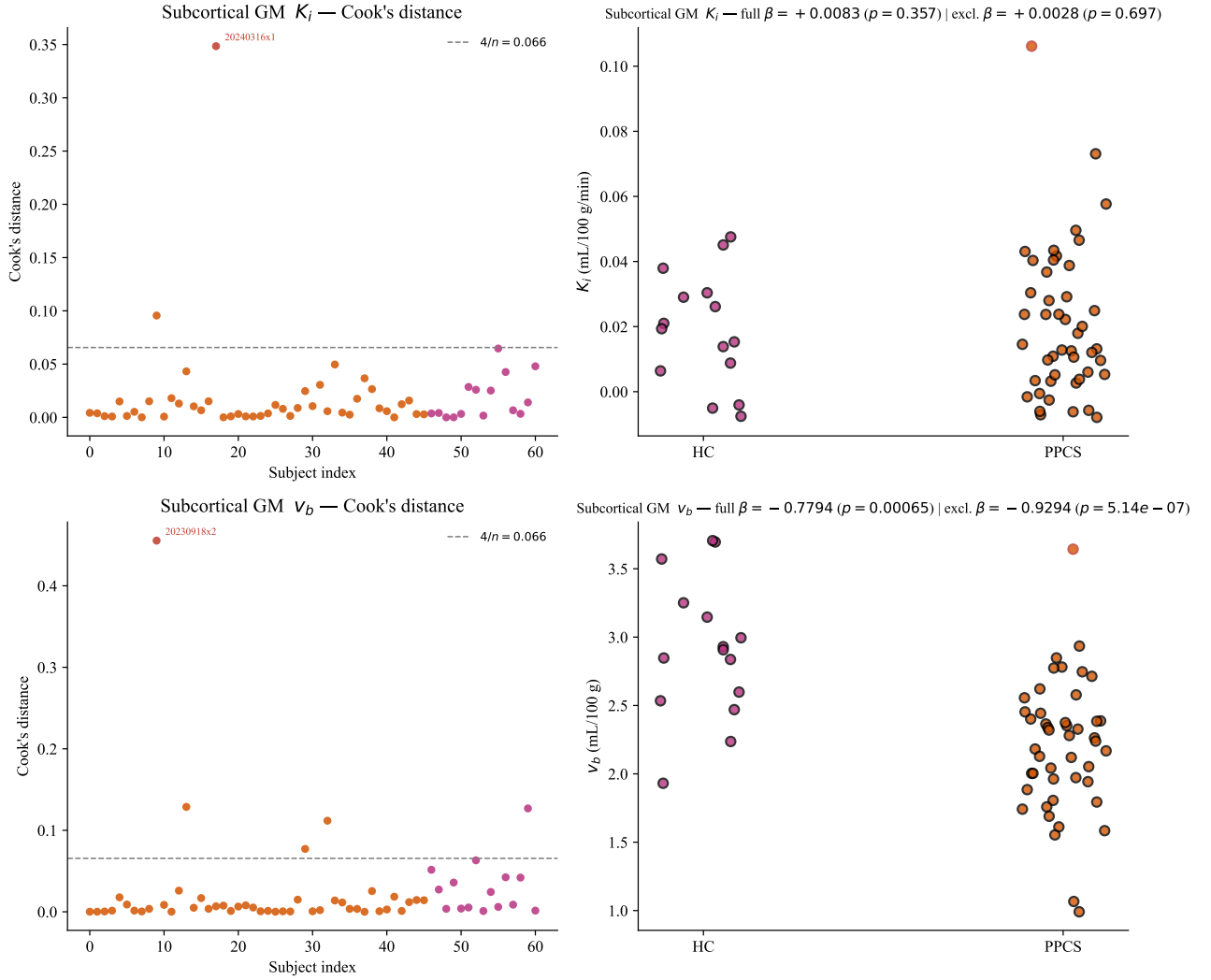


Supplementary Figure S1a: Group-mean parcel-level K_i and v_b in healthy controls ($n = 15$), shown on the same single-patient anatomical underlay used in the main parcel-difference figure (subject 20250217x4) for visual comparability. Per-metric colour scales are shared with the PPCS group map (Supplementary Figure S1b) so the two figures can be compared directly. Axial slices show only analysed parcel values; for visual continuity in the surface panels only, excluded cortical labels were coloured by neighbouring Desikan–Killiany surface parcels and remaining non-parcellated vertices were filled by vertex-neighbour interpolation, as in the main parcel-difference figure.



Supplementary Figure S1b: Group-mean parcel-level K_i and v_b in PPCS patients ($n = 46$), shown on the same single-patient anatomical underlay used in the main parcel-difference figure (subject 20250217x4) for visual comparability. Per-metric colour scales are shared with the healthy-control group map (Supplementary Figure S1a). Axial slices show only analysed parcel values; for visual continuity in the surface panels only, excluded cortical labels were coloured by neighbouring Desikan–Killiany surface parcels and remaining non-parcellated vertices were filled by vertex-neighbour interpolation, as in the main parcel-difference figure.

Supplementary Figure S2: Cook's distance for outlier-flagged tissue OLS models



Supplementary Figure S2: Sensitivity analysis for the two tissue-level OLS models flagged by the Bonferroni-corrected externally studentized residual screen: subcortical grey-matter K_i (top) and subcortical grey-matter v_b (bottom). Left: Cook's distance with the $4/n$ threshold; red markers and labels indicate the Bonferroni-significant observation. Right: individual values by group, with the flagged observation circled, and full vs. outlier-excluded HC3 coefficients shown in the panel title. The panel titles report raw HC3 p -values; the corresponding Benjamini–Hochberg FDR-corrected q -values (within each metric across the seven tissue compartments) are q_{FDR} from 0.625 to 0.990 for subcortical K_i (non-significant before and after exclusion) and from 9.10×10^{-4} to 3.8×10^{-6} for subcortical v_b (FDR-significant throughout), consistent with the main-text residual diagnostics.

Supplementary Table S1: Symptom-specificity regressions

PPCS-only associations between each tissue-level DCE parameter (K_i , v_b) and four symptom outcomes: Fatigue Severity Scale (FSS), and three Rivermead Post-Concussion Symptoms Questionnaire scores (RPQ-3 covering somatic items 1–3; RPQ-13 covering cognitive, emotional

and sensory items 4–16; and RPQ-total). Each row reports the age- and sex-adjusted OLS slope with HC3 standard errors for one tissue compartment, with FDR correction within each metric across the seven tissue compartments. Bold q entries indicate FDR significance ($q < 0.05$).

Metric	Tissue	FSS			RPQ-3			RPQ-13			RPQ-total		
		β	q		β	q		β	q		β	q	
K_i	Cortical GM	+29.76	0.948		-11.70	0.927		-6.45	0.884		-18.14	0.871	
K_i	Subcortical GM	+7.07	0.948		-19.09	0.927		-39.52	0.884		-58.61	0.871	
K_i	Brainstem	+35.18	0.948		-3.16	0.927		-11.81	0.884		-14.97	0.871	
K_i	Cerebellar GM	+5.98	0.948		-6.98	0.927		-35.37	0.884		-42.36	0.871	
K_i	White matter	+134.98	0.542		-1.44	0.927		+14.56	0.884		+13.12	0.871	
K_i	Cerebellar WM	+18.76	0.948		-7.17	0.927		-108.41	0.355		-115.58	0.597	
K_i	GM/WM interface	+90.62	0.948		-6.36	0.927		-7.70	0.884		-14.07	0.871	
v_b	Cortical GM	+1.73	0.896		-0.05	0.997		-0.54	0.852		-0.59	0.855	
v_b	Subcortical GM	+2.07	0.896		+0.08	0.997		-0.58	0.862		-0.50	0.908	
v_b	Brainstem	+1.10	0.896		-0.13	0.997		-2.20	0.756		-2.33	0.795	
v_b	Cerebellar GM	-0.26	0.896		-0.07	0.997		-1.78	0.756		-1.85	0.795	
v_b	White matter	+2.23	0.896		+0.00	0.997		-1.83	0.756		-1.83	0.795	
v_b	Cerebellar WM	+0.95	0.896		+0.11	0.997		-2.73	0.756		-2.61	0.795	
v_b	GM/WM interface	+1.98	0.896		-0.05	0.997		-1.24	0.756		-1.29	0.795	

Supplementary Table S2: Sensitivity analysis for parcel-level FDR-significant counts

Number of parcels reaching FDR significance ($q_{\text{FDR}} < 0.05$) for each metric under the primary parcel-level specification (raw OLS with HC3 standard errors, applied uniformly to all 85 parcels) and under a sensitivity specification that applies a uniform rank-inverse-normal (rank-INT) transformation of the outcome before fitting. Choroid-plexus, white-matter-hypointensity, banks of the superior temporal sulcus (bankssts), frontal-pole and temporal-pole labels (9 of 90 atlas parcels) were excluded from both analyses because they were not populated or had usable DCE coverage in fewer than 30% of subjects.

Metric	Raw OLS+HC3	Rank-INT
K_i	0/85	0/85
v_b	58/85	60/85

Supplementary Table S3: Per-parcel results

Per-parcel HC3 OLS group contrast (PPCS – HC, age + sex adjusted), Cohen’s d on raw means and Benjamini–Hochberg FDR-corrected q within each metric (85 tests per metric). Bold q entries indicate FDR-significance ($q < 0.05$).

Parcel	Lobe	K_i		v_b	
		d	q_{FDR}	d	q_{FDR}
Brainstem	Brainstem	−0.01	0.911	−1.06	0.002
L cerebellum WM	Cerebellum	−0.36	0.960	−1.26	0.001
L cerebellum cortex	Cerebellum	−0.17	0.730	−1.14	0.001
R cerebellum WM	Cerebellum	−0.47	0.957	−1.47	0.001
R cerebellum cortex	Cerebellum	−0.48	0.919	−1.22	<.001
L caudal anterior cingulate	Cingulate	−0.18	0.803	−0.65	0.069
L isthmus cingulate	Cingulate	+0.27	0.313	−1.33	<.001
L posterior cingulate	Cingulate	+0.21	0.312	−1.06	0.004
L rostral anterior cingulate	Cingulate	−0.02	0.404	−0.62	0.279
R caudal anterior cingulate	Cingulate	+0.23	0.763	−0.18	0.323
R isthmus cingulate	Cingulate	+0.15	0.167	−0.98	0.002
R posterior cingulate	Cingulate	+0.19	0.283	−0.63	0.019
R rostral anterior cingulate	Cingulate	+0.40	0.129	−0.41	0.289
L caudal middle frontal	Frontal	+0.18	0.911	−0.90	0.003
L lateral orbitofrontal	Frontal	+0.63	0.115	−0.06	0.289
L medial orbitofrontal	Frontal	−0.05	0.763	−0.08	0.425
L paracentral	Frontal	−0.43	0.960	−1.09	0.001
L pars opercularis	Frontal	+0.40	0.588	−0.93	0.014
L pars orbitalis	Frontal	−0.16	0.882	−1.46	<.001
L pars triangularis	Frontal	+0.20	0.679	−1.46	0.001
L precentral	Frontal	+0.61	0.129	−0.81	0.008
L rostral middle frontal	Frontal	+0.63	0.115	−0.93	<.001

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Parcel	Lobe	K_i		v_b	
		d	q_{FDR}	d	q_{FDR}
L superior frontal	Frontal	+0.65	0.129	-0.55	0.073
R caudal middle frontal	Frontal	-0.27	0.919	-0.44	0.107
R lateral orbitofrontal	Frontal	+0.40	0.313	+0.16	0.736
R medial orbitofrontal	Frontal	+0.61	0.129	-0.40	0.006
R paracentral	Frontal	-0.12	0.919	-0.59	0.003
R pars opercularis	Frontal	-0.05	0.313	-1.15	0.003
R pars orbitalis	Frontal	-0.11	0.919	-0.64	0.002
R pars triangularis	Frontal	-0.30	0.882	-1.69	<.001
R precentral	Frontal	+0.47	0.129	-0.76	0.032
R rostral middle frontal	Frontal	+0.24	0.305	-0.70	0.069
R superior frontal	Frontal	+0.24	0.804	-0.41	0.085
L insula	Insula	+0.11	0.897	-1.23	0.003
R insula	Insula	+0.05	0.336	-1.03	0.012
L cuneus	Occipital	+0.08	0.313	-1.43	<.001
L lateral occipital	Occipital	-0.05	0.882	-1.74	<.001
L lingual	Occipital	+0.60	0.129	-0.81	0.003
L pericalcarine	Occipital	+0.13	0.399	-1.05	0.015
R cuneus	Occipital	+0.09	0.313	-1.80	<.001
R lateral occipital	Occipital	-0.57	0.696	-1.50	<.001
R lingual	Occipital	+0.33	0.312	-1.11	0.004
R pericalcarine	Occipital	+0.23	0.356	-1.50	<.001
L inferior parietal	Parietal	+0.44	0.220	-1.53	<.001
L postcentral	Parietal	+0.29	0.630	-0.90	0.005
L precuneus	Parietal	+0.35	0.115	-1.01	<.001
L superior parietal	Parietal	+0.27	0.336	-1.66	<.001
L supramarginal	Parietal	+0.44	0.129	-1.21	<.001
R inferior parietal	Parietal	+0.29	0.313	-1.04	0.002
R postcentral	Parietal	+0.08	0.336	-0.82	0.049
R precuneus	Parietal	+0.40	0.115	-0.73	0.009
R superior parietal	Parietal	+0.26	0.404	-0.90	0.003
R supramarginal	Parietal	-0.15	0.803	-1.35	<.001
L thalamus	Subcortical	-0.50	0.897	-1.06	0.016
L caudate	Subcortical	-0.44	0.877	-0.31	0.343
L putamen	Subcortical	+0.23	0.348	-0.99	0.032
L pallidum	Subcortical	-0.08	0.882	-0.71	0.069
L hippocampus	Subcortical	-0.31	0.882	-0.92	0.002
L amygdala	Subcortical	-0.45	0.919	-1.25	0.006
L accumbens	Subcortical	-0.74	0.356	+0.10	0.979
L ventral DC	Subcortical	-0.20	0.882	-0.36	0.280
R thalamus	Subcortical	+0.39	0.882	-1.47	0.004
R caudate	Subcortical	+0.52	0.679	-1.02	0.004
R putamen	Subcortical	-0.12	0.851	-0.90	0.077
R pallidum	Subcortical	-0.11	0.877	-0.65	0.109
R hippocampus	Subcortical	-0.11	0.882	-0.82	0.016
R amygdala	Subcortical	+0.11	0.882	-0.82	0.005

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Parcel	Lobe	K_i		v_b	
		d	q_{FDR}	d	q_{FDR}
R accumbens	Subcortical	+0.24	0.851	-0.66	0.181
R ventral DC	Subcortical	+0.02	0.882	-0.42	0.321
L entorhinal	Temporal	+0.47	0.313	-0.55	0.131
L fusiform	Temporal	+0.45	0.313	-0.33	0.194
L inferior temporal	Temporal	+0.31	0.235	-0.60	0.131
L middle temporal	Temporal	+0.05	0.336	-1.80	<.001
L parahippocampal	Temporal	+0.17	0.803	-0.73	0.004
L superior temporal	Temporal	+0.24	0.226	-1.22	0.004
L transverse temporal	Temporal	+0.04	0.960	-1.18	0.010
R entorhinal	Temporal	+0.65	0.115	-0.77	0.066
R fusiform	Temporal	+0.36	0.313	-0.71	0.011
R inferior temporal	Temporal	+0.22	0.312	-0.55	0.073
R middle temporal	Temporal	-0.22	0.763	-1.54	<.001
R parahippocampal	Temporal	+0.48	0.313	-0.69	0.050
R superior temporal	Temporal	+0.37	0.312	-1.33	0.002
R transverse temporal	Temporal	+0.38	0.336	-0.79	0.069
L cerebral WM	WM	+0.29	0.313	-0.73	0.025
R cerebral WM	WM	+0.23	0.658	-0.44	0.111

Supplementary Table S4: MRI acquisition parameters

All sequences were acquired on a 3 T Philips Achieva dStream scanner (Philips Healthcare, Best, The Netherlands) with a 32-channel head coil. Structural sequences used compressed SENSE (factor 2); parameters are from the study acquisition protocol.

Sequence	TR/TE (ms)	Flip	FOV (mm ³)	Matrix	Slices
3D T1w TFE (structural)	11.13/5.11	8°	240 × 256 × 180	368 × 368	257
3D FLAIR	4800/340	90°	250 × 250 × 183	224 × 223	326
3D T2w	3000/280	90°	250 × 250 × 180	252 × 251	360
SWI (6-echo)	51/multi-echo	20°	230 × 140 × 163	384 × 384	140
Pre-contrast T1 map (SR)	3.95/1.88	30°	230 × 94 × 183	96 × 75	10
DCE (2D TurboFLASH)	3.95/1.88	30°	230 × 94 × 183	96 × 75	10

The pre-contrast T1 map used seven saturation delays (TS = 120, 300, 600, 1000, 2000, 4000, 10 000 ms); its geometry matched the DCE series. The DCE series comprised 250 dynamic frames at 2.44 s temporal resolution (acquired in-plane voxel $2.4 \times 2.4 \text{ mm}^2$, reconstructed to $0.90 \times 0.90 \text{ mm}^2$).

Supplementary Table S5: Fazekas scores

Individual Fazekas ratings for the 46 patients with PPCS in the DCE-MRI analysis cohort, shown as the two component subscores: periventricular white matter (PV, 0–3) and deep white matter and subcortical regions (Deep, 0–3). Healthy controls were not part of the trial’s structural-imaging review and are not included here. Patients are numbered sequentially and do not correspond to internal study identifiers. A dash (–) indicates a rating not available. One patient (P43) had a single microbleed in the left frontal region on susceptibility-weighted imaging; no other microbleeds were found in this cohort.

Patient	PV	Deep	Patient	PV	Deep	Patient	PV	Deep
P1	0	0	P17	1	0	P33	0	0
P2	0	0	P18	1	2	P34	0	2
P3	0	0	P19	0	1	P35	1	2
P4	0	1	P20	0	0	P36	0	0
P5	0	0	P21	0	1	P37	0	1
P6	0	0	P22	0	0	P38	1	2
P7	0	0	P23	1	1	P39	0	0
P8	–	–	P24	1	0	P40	0	0
P9	0	0	P25	0	0	P41	1	0
P10	0	0	P26	0	0	P42	1	1
P11	0	0	P27	1	0	P43	1	1
P12	0	0	P28	0	1	P44	1	0
P13	0	0	P29	0	0	P45	1	1
P14	0	0	P30	0	1	P46	1	0
P15	1	1	P31	0	0			
P16	0	0	P32	0	0			

Supplementary Appendix A: Capillary-radius plausibility calculation

This calculation is a simplified model used only to test whether the magnitude of the observed v_b reduction is numerically compatible with the lower oxygen extraction inferred in the related CMRO₂ study. It should not be read as proof of capillary narrowing.

Model a capillary as a circular cylinder of radius r and length L . A bed of N such capillaries has blood volume V_b and oxygen-exchange surface area S ,

$$V_b = N\pi r^2 L, \quad S = N \cdot 2\pi r L. \quad (1)$$

The blood volume is proportional to the cross-sectional area of the lumen, whereas the exchange surface is proportional to the circumference of the capillary wall. Thus, if N and L are fixed,

$$V_b \propto r^2, \quad S \propto r. \quad (2)$$

Oxygen extraction can be written using the Crone–Renkin relation [1, 2],

$$E = 1 - e^{-PS/f}, \quad (3)$$

where P is the permeability to oxygen, S is exchange surface area, and f is perfusion. Comparing controls (subscript 1) and PPCS patients (subscript 2), and assuming equal P and f , the extraction fractions imply the surface-area ratio

$$\frac{S_2}{S_1} = \frac{\ln(1 - E_2)}{\ln(1 - E_1)}. \quad (4)$$

If the group difference is driven only by capillary radius, then $r_2/r_1 = S_2/S_1$, and the corresponding blood-volume ratio is

$$\frac{V_{b,2}}{V_{b,1}} = \left(\frac{r_2}{r_1}\right)^2 = \left(\frac{S_2}{S_1}\right)^2 = \left[\frac{\ln(1 - E_2)}{\ln(1 - E_1)}\right]^2. \quad (5)$$

In the related CMRO₂ study, oxygen extraction fractions were approximately $E_1 = 0.32$ in controls and $E_2 = 0.28$ in PPCS patients [3]. This gives

$$\frac{S_2}{S_1} = \frac{\ln(1 - 0.28)}{\ln(1 - 0.32)} = \frac{\ln 0.72}{\ln 0.68} \approx 0.852. \quad (6)$$

Under the radius-only assumption, this corresponds to an approximately 15% smaller capillary radius. Because volume scales with the square of radius,

$$\frac{V_{b,2}}{V_{b,1}} = 0.852^2 \approx 0.726, \quad (7)$$

or an approximately 27% reduction in exchanging capillary blood volume. This is of the same order as the approximately 20% reduction in DCE-derived v_b observed in the present study.

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